

SWEEP rule on the KD-Toric code

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1 Tree Separator Lemma

For a tree $T = (V, E)$ and a vertex $v \in V$, we call the subtree obtained by taking the descendants of v the subtree rooted at v .

Claim 1.1. Let $T = (V, E)$ be a tree with $m > 3$ leaves and τ edges, then it has a subtree with $m' \in (\frac{1}{3}m, \frac{2}{3}m)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Proof. Order the vertices according to their heights. Denote by $v \in V$ the lowest vertex for which the subtree rooted at v has at least $\frac{1}{3}m$ leaves. Now, denote by T_1 the subtree obtained through the following iterative process over v 's children:

1. Initialize T_1 as the subtree rooted at v .
2. For a child u of v :
 - (a) Check if erasing from T_1 the subtree rooted at u doesn't decrease the number of leaves below $\frac{1}{3}m$. If so, then we update T_1 to be the result of the erasure.

Clearly, the algorithm is defined such that T_1 has to have at least $\frac{m}{3}$ leaves. Furthermore, since any child of v is a root of a subtree with fewer than $\frac{m}{3}$ leaves (otherwise we get a contradiction to the minimality of v 's height), we have that if T_1 would have more than $\frac{2}{3}m$ leaves, then the subtraction of any of its children would yield a tree with at least $\frac{2}{3}m - \frac{1}{3}m = \frac{1}{3}m$ leaves. Namely, v has in T_1 a child u such that the subtraction of the subtree rooted at u from T_1 results in a subtree with more than $\frac{1}{3}m$ leaves. But if indeed there were such a child, then the algorithm would erase it, so there is not.

Now let $T_2 = T/T_1 \cup v$, namely the tree induced by the subtraction of T_1 's edges from T . Observe that since the summation of leaves in T_1 and T_2 is the leaves in T , and T_1 has no more than $\frac{2}{3}m$ leaves, and no less than $\frac{1}{3}m$ leaves, it holds that the number of leaves in T_2 is also in the range $(\frac{1}{3}m, \frac{2}{3}m)$. We claim that at least one of the subtrees has fewer than $\frac{m'}{m}\tau$ edges.

To see this, denote by m_1, m_2 and by τ_1, τ_2 the leaves and the edges in T_1, T_2 respectively. Now:

$$\frac{\tau}{m} = \frac{\tau_1 + \tau_2}{m_1 + m_2} = \frac{m_1}{m_1 + m_2} \frac{\tau_1}{m_1} + \frac{m_2}{m_1 + m_2} \frac{\tau_2}{m_2}$$

Namely, $\frac{\tau}{m}$ is a weighted average of $\frac{\tau_1}{m_1}$ and $\frac{\tau_2}{m_2}$, and therefore it is greater than their minimum. Peak the subtree with the minimal quotient. $\square$

By repeating recursively on the above construction, we get the following corollary:

Corollary 1.2. Let $T = (V, E)$ be a tree with m leaves and τ edges. Then, for any $\alpha > 0$, it has a subtree with $m' \in (\frac{1}{3}\alpha m, \frac{2}{3}\alpha m)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Claim 1.3. Let $\gamma, \beta \in \mathbb{R}_{>0}$ and n be a positive integer. For any tree T with $\tau \geq \beta n$ edges and m leaves such that $\tau \leq \gamma m$, there is a subtree T' of T with a number of edges τ' less than $\frac{\beta}{2}n$ and a number of leaves greater than $\frac{\beta}{4\gamma}n$.

Proof. Since the construction of claim 1.1 preserves the ratio of edges to leaves, we have that for a T' obtained using the construction, the number of edges satisfies $\tau' \leq \gamma m'$. So, it's enough to look for the maximal α such that:

$$\gamma m' \leq \gamma \frac{2}{3}\alpha m \leq \frac{\beta}{2}n \Rightarrow \alpha = \frac{3\beta}{4\gamma} \frac{n}{m}$$

Which gives $m' \geq \frac{1}{3}\alpha m = \frac{\beta}{4\gamma}n$.

$\square$